

## MAT 426, Class 3: Gaussian Elimination with Partial Pivoting

### Practical Consideration: Storage of $L$ and $U$ in Algorithm

How many flops are required for finding LU decomposition using Gaussian Elimination?

*Note:* Gaussian elimination can fail because a \_\_\_\_\_ can become \_\_\_\_\_.

**Example 1.** Consider the matrix  $A = \begin{pmatrix} 0.0001 & 1 \\ 1 & 1 \end{pmatrix}$ .

- (a) Use Gaussian elimination with three digit arithmetic to find the  $LU$  decomposition of  $A$ .
  
  
  
  
  
  
  
  
  
  
- (b) Use Gaussian elimination using three digit arithmetic but allowing a row interchange to find the  $LU$  decomposition of  $A$ .

## Gaussian Elimination with Partial Pivoting

*Goal:* Use \_\_\_\_\_ to write the matrix  $A$  as \_\_\_\_\_.

*Why is this factorization important?*

**Definition.** A nonzero square matrix  $P$  is called a permutation matrix if there is exactly \_\_\_\_\_ nonzero entry in each \_\_\_\_\_ and \_\_\_\_\_ column which is \_\_\_\_\_ and the rest are all zero.

**Example 2.** Consider a  $3 \times 3$  matrix  $A$  and the permutation matrix  $P_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ .

(a) Compute  $P_1 A$

(b) Compute  $AP_1$

*Note:*

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*Some Important Ingredients:*

- For a permutation matrix  $P$ ,  $P^2 =$  \_\_\_\_\_. Why? Permutation matrices are \_\_\_\_\_.

**Example 3.** Let  $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ . Compute  $P^2$ .

- For a permutation matrix  $P$  and an elementary matrix  $M$ ,  $PMP =$  \_\_\_\_\_, where \_\_\_\_\_ is an \_\_\_\_\_ matrix.

**Example 4.** Compute the following product:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 3/4 & 1 & 0 \\ -1/4 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} =$$

- The product of two permutation matrices  $P_1$  and  $P_2$  is a \_\_\_\_\_ matrix.

**Example 5.** Compute the following product:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} =$$

### Gaussian Elimination with Partial Pivoting (GEPP)

Let  $A^{(0)} = A$ . Then, to obtain  $A^{(k)} = (a_{ij}^{(k)})$ :

1.

2.

3.

4.

**Example 6.** Use Gaussian Elimination with partial pivoting to row-reduce  $A = \begin{pmatrix} 1 & 7 & -3 \\ -3 & 2 & 0 \\ 4 & -1 & 3 \end{pmatrix}$  to an upper-triangular matrix.

### Extracting $P$ , $L$ to get $PA = LU$

**Example 7.** Returning to Example 1, determine the  $LU$  decomposition of  $A = \begin{pmatrix} 1 & 7 & -3 \\ -3 & 2 & 0 \\ 4 & -1 & 3 \end{pmatrix}$

(a) How do we approach this for any  $3 \times 3$  matrix?

(b) What is the  $PA = LU$  decomposition for the original matrix  $A$ ?